CAMPUS ALERT – College Event Notification App

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled "Campus Alert" is the Bonafide work of "JANANY M (2116231901012) & P LAKSHMI PRIYA(2116231901025) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

CampusAlert is a mobile application developed to improve communication and awareness of college events among students. In many institutions, event information is often shared through notice boards, emails, or messaging platforms, which can lead to delayed communication and missed opportunities. To address this issue, the proposed system provides a centralized platform for delivering real-time event notifications directly to students' mobile devices.

The application is developed using Kotlin for Android and integrates Firebase services for authentication, database management, and push notifications. The system allows administrators to create, update, and manage event details such as date, time, and venue, while students can view upcoming events and register through the application. Notifications are automatically sent to users whenever new events are posted, ensuring timely updates.

The main objective of CampusAlert is to enhance campus communication, increase student participation in events, and provide a user-friendly interface for efficient event management. The system offers a reliable and scalable solution that simplifies the process of sharing and accessing event information within a college environment.

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CHAPTER 1
INTRODUCTION

1.1 GENERAL

In modern educational institutions, effective communication is essential for the smooth conduct of academic and extracurricular activities. Colleges organize various events such as seminars, workshops, technical fests, and cultural programs. However, the information regarding these events is often communicated through traditional methods like notice boards, emails, or social media groups, which may not be efficient or reliable.

Due to this, students may miss important announcements or receive information late. With the increasing use of smartphones, there is a need for a digital solution that can provide quick and centralized access to event-related information. The proposed system, **CampusAlert**, is a mobile application designed to deliver real-time notifications and updates about college events directly to students.

1.2)SCOPE OF THE WORK

The scope of this project is to design and develop a mobile application that improves communication between students and event organizers. The system allows administrators to create, manage, and update event details such as date, time, and venue.

Students can view upcoming events, receive instant notifications, and register for events through the application. The system uses a centralized database to store and manage all information efficiently. The application is scalable and can be implemented in different educational institutions to enhance event management and student engagement..

1.2 AIM AND OBJECTIVES OF THE PROJECT

The main aim of this project is to develop a mobile application named CampusAlert that serves as a centralized platform for delivering real-time notifications and information about college events. The system is designed to enhance communication between students and event organizers by providing timely updates and easy access to event details. It ensures that students stay informed about all academic and extracurricular activities and do not miss important opportunities.

The objectives of the project include designing and developing a user-friendly Android application using Kotlin that provides a smooth and efficient user experience. The system aims to deliver real-time notifications to students whenever new events are created or updated. It enables administrators to efficiently create, modify, and manage event details such as date, time, and venue. The application allows students to view upcoming events, access detailed information, and register for events directly through the platform.

Additionally, the project focuses on maintaining a centralized and secure database to store event and user information. It aims to simplify the process of event communication, reduce dependency on traditional methods, and improve overall coordination within the institution. Ultimately, the system seeks to increase student engagement and participation in college activities by making event information easily accessible and readily available.

CHAPTER 2
LITERATURE SURVEY

The development of mobile applications for event management and notification systems has gained significant attention in recent years. Various systems have been proposed to improve communication and ensure timely delivery of information in educational institutions.

A study on mobile-based event management systems highlights the importance of centralized platforms for sharing event information. These systems allow users to access event details, schedules, and updates through a single interface, reducing dependency on traditional communication methods such as notice boards and emails.

Another research work focuses on the use of push notification services in mobile applications. It emphasizes that real-time notifications improve user engagement and ensure that important information reaches users instantly. Technologies such as Firebase Cloud Messaging are widely used for this purpose due to their efficiency and reliability.
Existing college management systems provide features such as attendance tracking, timetable management, and announcements. However, many of these systems lack a dedicated and efficient event notification module, which limits their effectiveness in managing campus events.

Some event management applications also include features like online registration and user authentication, which simplify the process for both organizers and participants. These systems demonstrate the need for secure login mechanisms and centralized data storage.
Based on the analysis of existing systems, it is observed that there is a need for a simple, user-friendly, and efficient mobile application that focuses specifically on event notifications and management. The proposed **CampusAlert** system aims to address these gaps by providing real-time alerts, easy event access, and a centralized platform for both students and administrators.

CHAPTER 3
SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

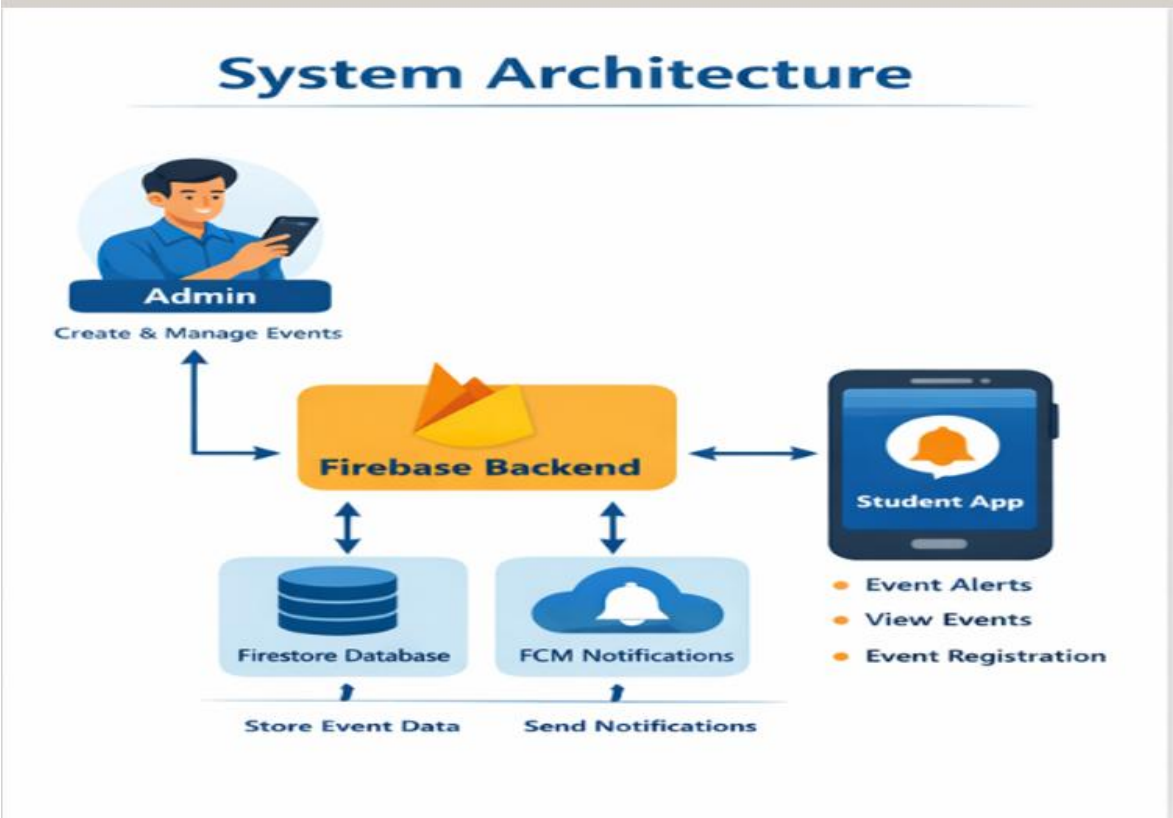


Fig 3.1.1 – Architecture Diagram

The architecture of the CampusAlert system is designed as a client-server model that ensures efficient communication between users and the database.

The system consists of three main components: the Android mobile application (client side), the backend services, and the cloud database. The mobile application is developed using Kotlin and provides the user interface for both students and administrators.

Administrators use the application to create and manage event details such as event name, date, time, and venue. These details are stored in the cloud database using Firebase Firestore. When a new event is added or updated, the backend service triggers notifications using Firebase Cloud Messaging (FCM).

Students access the application to view event details and register for events. Whenever a new event is posted, students receive instant notifications on their mobile devices. The system ensures real-time synchronization between the database and the application.

The architecture provides a scalable, secure, and efficient solution for managing and delivering event notifications within a college environment.

3.2 USE CASE DIAGRAM

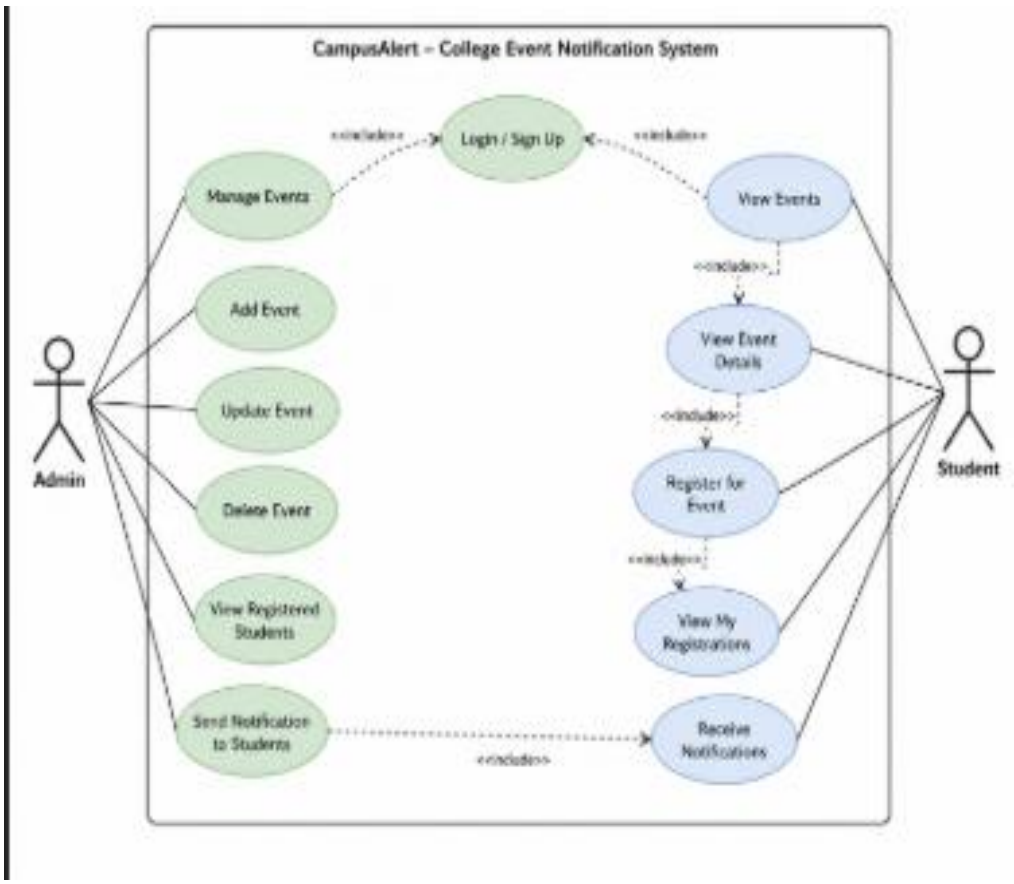


Fig 3.2.1 – Use Case Diagram

The Use Case Diagram of the **CampusAlert** system represents the interaction between users and the application. The system mainly involves two actors: **Admin** and **Student**.

The **Admin** is responsible for managing the events in the system. The admin can log in to the application, add new events, update existing events, and delete events when necessary. The admin can also view the list of registered students for each event and send notifications to inform students about new or updated events.

The **Student** interacts with the system by logging in to the application and accessing event-related information. Students can view all available events, check detailed information about each event, and register for events through the application. They also receive notifications whenever new events are announced or updated.

The use case diagram clearly shows how both admin and student interact with different functionalities of the system. It helps in understanding the overall working of the application and the roles of each user in the system.

3.3 HARDWARE SPECIFICATIONS

The **CAMPUS ALERT** mobile app is designed to function smoothly on standard smartphones and computing devices, without requiring specialized hardware. The following hardware specifications are recommended for optimal performance:

3.3.1 **Processor**: Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher, MediaTek Helio P60 or higher)

3.3.2 **RAM**: Minimum 3 GB (Recommended: 4 GB for smoother performance)

3.3.3 **Storage**: 32 GB internal storage or higher (to store user data and app files)

3.3.4 **Graphics**: Integrated graphics (no dedicated GPU required)

3.3.5 **Operating System**: Android 8.0 (Oreo) or higher

3.3.6 **Other Requirements**: Stable internet connection for cloud synchronization and data storage, touchscreen support, optional hardware for voice assistant features.

These specifications ensure the app runs efficiently, allowing smooth task management, note-taking, and real-time features while providing a seamless user experience across a variety of devices.

3.4 SOFTWARE SPECIFICATIONS

The development of the **CampusAlert** mobile application requires specific software tools and technologies to ensure proper functionality and performance.

3.4.1 Operating System

- Windows 10 / Windows 11
  (Used for developing and testing the application)

3.4.2 Programming Language

- Kotlin
  (Used for developing the Android application)

3.4.3Development Tool

- Android Studio
  (Integrated Development Environment for building the application)

3.4.4 Database

- Firebase Firestore
  (Cloud-based database for storing event and user data)

3.4.5Backend Services

- Firebase Authentication
  (Used for user login and authentication)

3.4.6Notification Service

- Firebase Cloud Messaging (FCM)
  (Used to send push notifications to users)

3.4.7 Other Tools

- XML (for UI design)
- Git (optional, for version control)

CHAPTER 4
PROPOSED SYSTEM

The proposed system, CampusAlert, is a mobile application designed to provide a centralized and efficient platform for managing and delivering college event notifications. The system aims to overcome the limitations of traditional communication methods by offering real-time updates and easy access to event information.

In this system, administrators can create, update, and manage event details such as event name, date, time, venue, and description through the mobile application. These details are stored in a cloud-based database, ensuring that information is always up to date and easily accessible.

Students can log in to the application to view all upcoming events in a structured and user-friendly interface. They can access detailed information about each event and register directly through the app. Whenever a new event is created or an existing event is updated, students receive instant notifications on their mobile devices.

The proposed system uses modern technologies such as Kotlin for Android development and Firebase services for database management, authentication, and notifications. This ensures that the system is scalable, secure, and efficient.

Overall, the CampusAlert system provides a reliable solution for improving communication, increasing student participation, and simplifying event management within a college environment.

CHAPTER 5
MODULE DESCRIPTION

The CampusAlert system is divided into several modules, each responsible for specific functionalities of the application.

1 User Authentication Module

This module handles the login and registration process for both students and administrators. It ensures secure access to the system using valid credentials and prevents unauthorized access.

2 Event Management Module

This module is used by the admin to create, update, and delete event details such as event name, date, time, venue, and description. It helps in efficiently managing all campus events.

3 Event Notification Module

This module is responsible for sending real-time notifications to students whenever a new event is added or updated. It ensures that users receive instant alerts on their mobile devices.

4 Event Viewing Module

This module allows students to view all available events in a structured format. Students can access complete event details, including date, time, and venue.

5 Event Registration Module

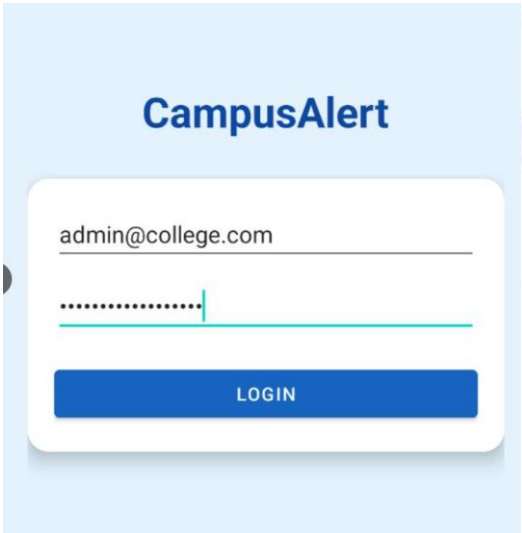
This module enables students to register for events directly through the application. It records the registration details and helps organizers track participant information.

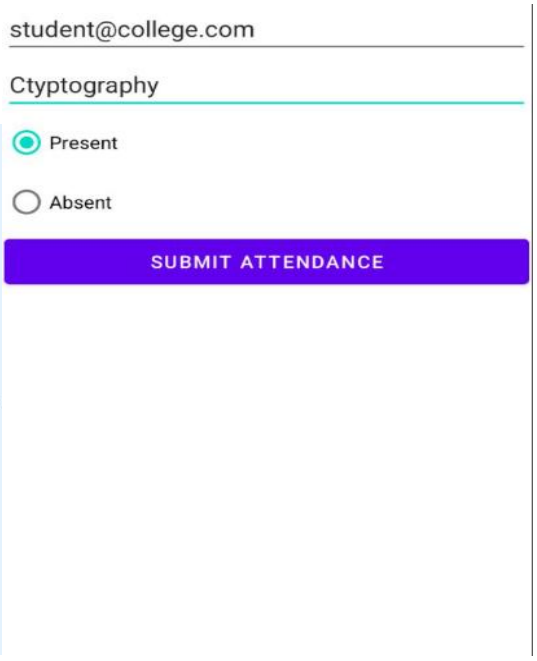
6 Database Management Module

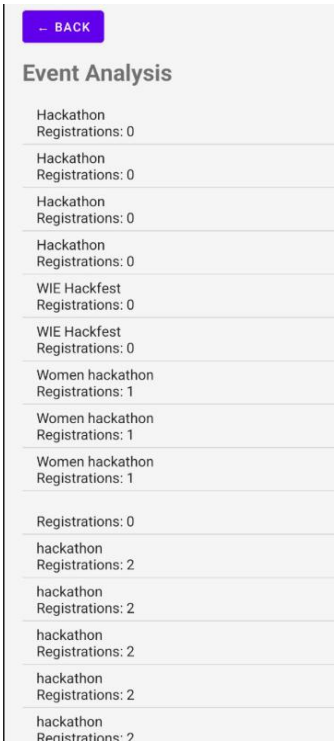
This module manages the storage and retrieval of data such as event details, user information, and registrations. It ensures data consistency and secure handling using a cloud database.

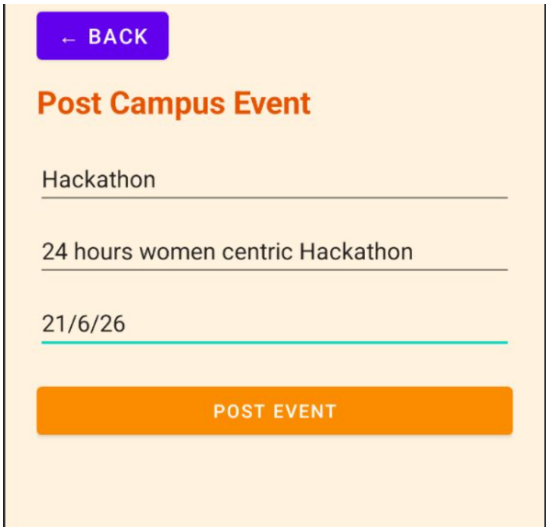
CHAPTER-6

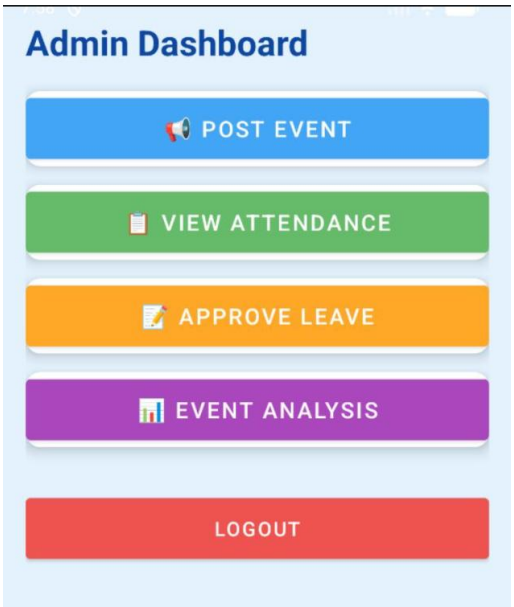
OUTPUT

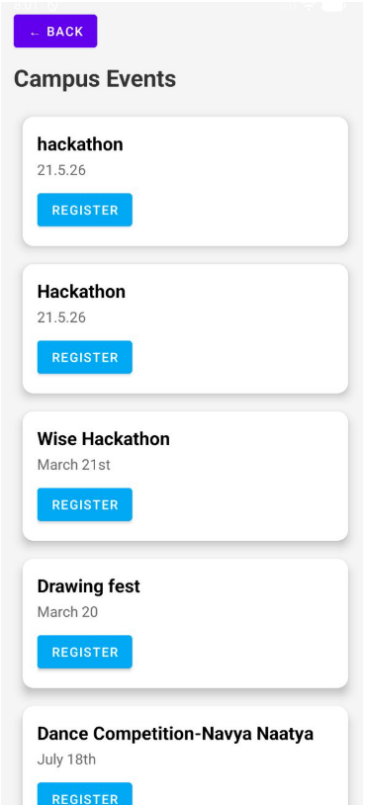


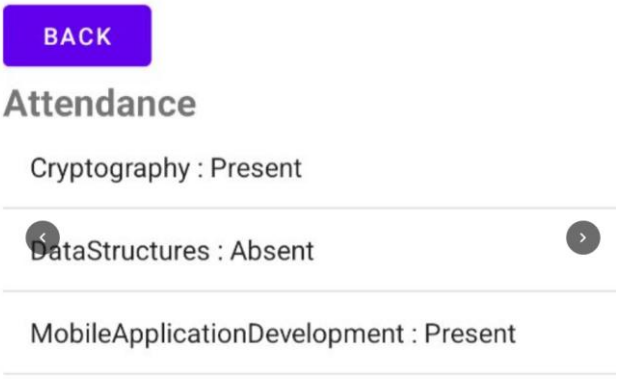


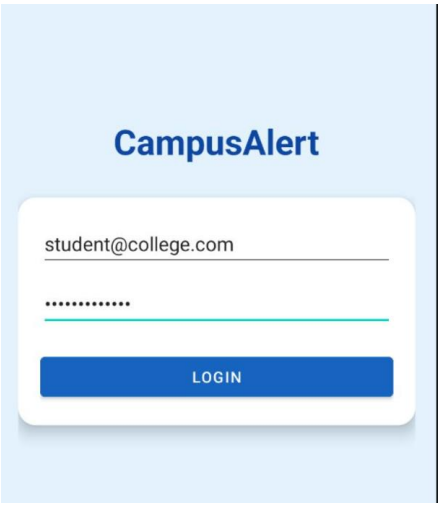





Admin Dashboard
POST EVENT
VIEW ATTENDANCE
APPROVE LEAVE
EVENT ANALYSIS
LOGOUT


BACK
Campus Events
hackathon
21.5.26
REGISTER
Hackathon
21.5.26
REGISTER
Wise Hackathon
March 21st
REGISTER
Drawing fest
March 20
REGISTER
Dance Competition-Navya Naatya
July 18th
REGISTER


BACK
Attendance
Cryptography : Present
DataStructures : Absent
MobileApplicationDevelopment : Present


CampusAlert
student@college.com
LOGIN

CHAPTER 7
CONCLUSION AND FUTURE ENHANCEMENTS

The **CampusAlert** mobile application provides an efficient and reliable solution for managing and delivering college event notifications. It eliminates the limitations of traditional communication methods by offering a centralized platform for sharing event information. The system ensures that students receive real-time updates, making it easier for them to stay informed about campus activities.

The application is user-friendly and allows both administrators and students to interact with the system effectively. Administrators can manage events easily, while students can view and register for events without difficulty. Overall, the system improves communication, saves time, and increases student participation in college events.

Future Enhancements:

The proposed system can be further improved by adding the following features:

- **AI-Based Event Recommendations**
  Suggest events to students based on their interests and previous participation.
- **QR Code Attendance System**
  Enable event check-in using QR codes for faster and accurate attendance tracking.
- **Event Calendar Integration**
  Allow students to sync events with their personal calendar.
- **Feedback and Rating System**
  Collect student feedback and ratings for events.
- **Multi-Platform Support**
  Extend the application to iOS and web platforms.
- **Advanced Analytics for Admin**
  Provide reports on event participation and engagement.

CHAPTER 8

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